

EDUCATION

Sharif University of Technology

M.S. in Geotechnical Engineering

Thesis: Evaluation of Dynamic Properties of Plastic Concrete Using Bending Disk Tests in Small Strain Range

Tehran-Iran

2008-2010

Water and Power Industry College of Shahid Beheshti University

B.S. in Dam and Water Networks Engineering

Thesis: Application of HEC-FDA for Economic Analysis of Flood Management Projects in Gorganrod River

Tehran-Iran

2004-2008

PROFESSIONAL EXPERIENCE

ABFAN Company (Grade 1 in Dam engineering)

2010-2023

- Project Manager (TangeHamam Dam, Goharkuh Dam, Boroujerdi Dam)
- Embankment Dam designing (Galal Dam, Saghaveh Dam, DashtePalang Dam)
- Cemented Materials Dam Design (DashtePalang Dam)
- Embankment Dam monitoring (SefidRoud Dam, Gavoshan Dam, Soleymanshah Dam, GilaneGharb Dam, Shian Dam, Hale Dam, Golabar Dam, Kinevars Dam, Marvak Dam)
- Dynamic analysis of Embankment dams (QasreShirin Dam, Galal Dam, TangeHamam Dam)
- Post-earthquake seismic evaluation of Tang Hamam Dam
- Tunnel Designing (Patagh Tunnel, Kahir Tunnel)
- Reinforced slope designing (Many Projects)
- Numerical Modeling (Many Projects)

GEOLAB Company (Startup)

2023-Present

- Founder
- Providing Internet of Things services in civil engineering
- Providing structural health monitoring services
- Providing machine learning and image processing services in civil engineering
- Manufacturing various types of measurement sensors and data loggers in civil engineering

AZMOON Company (Knowledge-based)

2015- Present

- Research and Development Manager
- Product Manager of advanced laboratory equipment including bender element, ultrasonic, concrete 3D printer, rapid chloride penetration, rebar tensile, resonant frequency, concrete maturity, impact response, pulse response and pile dynamic test, pile integrity test, etc.
- Programming of web, Windows and Android applications
- Programming of microcontrollers
- Mastery of industrial hardware and software protocols
- Mastery of civil codes including: USBR, USACE, ICOLD, ASCE and publications of the program organization
- Mastery of soil, rock and concrete testing standards including National Standard, ASTM, AASHTO, EN and ISRM

MOSHANIR Company (Grade 1 in Dam engineering)

2022- present

- Geotechnical expert (Lar Dam stability study)
- Embankment dams monitoring

OTHER EXPERIENCE

- Dams monitoring in Kermanshah province (Gavoshan, Soleimanshah, Gilanegharb and shian), Zanjan province (Taham, Golaber and Kineh Vars) Lorestan province (Hale, Koznar and Khan Abad)
- Investigating the problems of Sefidrood Dam instrumentation and providing solutions

- Holding numerous training workshops in the field of structural health monitoring, dam monitoring, CMD design, earthquakes engineering, non-destructive testing
- Cooperation in the review Guide publications for dam foundation improvement, Embankment dam design
- Research on the efficiency of filters under different erosion conditions in Embankment dams in the Advanced Soil Mechanics Laboratory of the University of Tehran
- Evaluation of dynamic parameters of plastic concrete (Master's thesis)
- Research on the application of uncertainty and risk in civil engineering
- Economic comparison of flood management projects (Bachelor's thesis)

SOFTWARE SKILLS

1. Programing language

C++ C# Python HTML Java

2. Geotechnical Engineering

-Geostudio -Itasca (FLAC, UDEC) -MIDAS GTS NX -Rocscinse (Phase2, Slide, RocPlane, RocSupport, Settle3D, Swedge, Unwedge, RocData, Dips) -GEO5 -PLAXIS -SHAKE -RockWork

3.General

-Global Mapper (GIS) -AutoCAD -Civil3D -Photoshop -InDesign -MS Office -MS Project

Language: English

Reading: fluent Writing: fluent Listening: moderate Speaking: moderate (IELTS 6.5)

PUBLICATION

- 1-GhezelAyagh A., Mohammadian M., Fadaei M., Araghian H., 2017, Upstream face of CMDs (Case Study: first CMDs in IRAN), 85th Annual Meeting of ICOLD, Prague, Czech Republic.
- 2-GhezelAyagh A., Mohammadian M., Fadaei M., 2017, CMDs on low modulus foundations, Four first experiences in Iran, 85th Annual Meeting of ICOLD, Prague, Czech Republic.
- 3-Fadaei M., Jafarzadeh, F., (in press), Evaluation of Dynamic Properties of Plastic Concrete Using Bending Disk Tests in Small Strain Range.
- 4-Ghezelayagh A., Fadaei M., Bolouri A., 2017, Long-term behavior of the Kinevars embankment dam using a numerical model and compare it with instrumentation results, 4th International Conference on Long-Term Behavior and Environmentally Friendly Rehabilitation Technologies of Dams (LTBD 2017), Tehran, Iran.
- 5-Fadaei M., Razmkhah M., 2018, ISO 9001:2015 Interpretation and Application Guide 1st edition 2017 (Persian Translation)
- 6-Fadaei M., Razmkhah M., 2018, ISO 9004:2018 Quality management - Quality of an organization - Guidance to achieve sustained success (Persian Translation)
- 7-Fadaei M., Akbari Paydar N., Razavi F., Mohammadian A., 2016, Control of erosion at slopes of Tang-e-Hammam channels using Geocell, 5th International Conference on Geotechnical Engineering and Soil Mechanics, Tehran, Iran. (in Persian)
- 8-Akbari Paydar N., Bolouri A., Fadaei M., 2011, Studies on Water Pressure Changes in Different Parts Kinevars Dam During Construction and First Impounding, 6th National Congress on Civil Engineering, Semnan, Iran. (in Persian)
- 9-Ahmadi M.M., Akbari Paydar N., Fadaei M., 2011, Measurement of compressive and shear velocity using Bender Element & Piezoelectric Disc, first international conference of urban construction in the vicinity of active faults, Tabriz, Iran. (in Persian)
- 10-Jafarzadeh F., Fadaei M., Akbari N., 2011, Application of piezoelectric in seismic geotechnical engineering and measurement of compressive wave velocity using piezoelectric, 6th International Conference on Seismology and Earthquake Engineering, Tehran, Iran. (in Persian)
- 11-Fadaei M., 2010, stability analysis of left abutment of Latian dam by calculation of reliability coefficient using Montecarlo method, 4th International Conference on Geotechnical Engineering and Soil Mechanics, Tehran, Iran. (in Persian)
- 12-Fadaie M., Attari J., Mesbahi M.M., 2009, Economic Comparison of Flood Management Projects in the Gorganrod River, 8th International Congress on Civil Engineering, Shiraz, Iran. (in Persian)
- 13-Akbari Paydar N., Bolouri A., Fadaei M., 2011, Behavior of the Kinevars Dam at the time of construction and operation by studying changes in water pressure in the body and the foundation, the first international conference of dam and hydroelectric power plant, Tehran, Iran. (in Persian)

- 14-Akbari Paydar N., Fadaei M., 2012, Evaluation of Critical Hydraulic Gradient of Cohesive Soils Using Experimental and Analytical Methods, 9th International Congress on Civil Engineering, Isfahan, Iran. (in Persian)
- 15-Fadaei M., Narue A., 2015, Technical and Economic Review of Sealing Elements for Goharkuh dam, 14th Hydraulic Conference of Iran, Zahedan, Iran. (in Persian)
- 16-Fadaei M., Akbari Paydar N., Razavi F., Mohammadian A., 2016, Studies on changes in water pressure in different parts of body and body during construction and first drainage of Tang-e-Hamam Dam, fourth dam and tunnel conference, Tehran, Iran. (in Persian)
- 17-Fadaei M., Mohammadian A., Akbari Paydar N., 2016, Combined method of sealing face in FSHD dams (Case study in Kahir and dashtepalang dams), 4th Dams and Tunnels Conference, Tehran, Iran. (in Persian)